REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Picturing Motion Practice Questions

Grade 9 Advanced | Module 2 Lesson 1 | Picturing Motion

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Equal-time motion diagrams: compare successive positions and their spacing. Elapsed time = number of intervals × interval duration.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 29–31.

P01. An object is moving relative to a reference point when its:

A. Mass changes B. Position changes with time
C. Colour changes D. Temperature is constant

P02. A particle model represents an object by:

A. Its detailed surface texture B. Only its mass
C. A point at each recorded position D. A continuous photograph without timing

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P03. In a motion diagram, successive images should usually be separated by:

A. Equal distances regardless of motion B. Random times
C. Different coordinate units D. Equal time intervals

P04. Equal dot spacing along a straight path suggests:

A. Uniform motion B. Speeding up
C. Slowing down D. An instant change of direction

P05. Increasing dot gaps at equal time intervals indicate:

A. Decreasing speed B. Increasing speed
C. No motion D. A smaller elapsed time between every image

P06. A motion diagram shows positions 0, 4, 8 and 12 m at equal times. What is the spacing between successive positions?

A. 8 m B. 2 m
C. 4 m D. -4 m

P07. A camera records 5 successive positions separated by 0.2 s. Calculate the time from the first image to the last.

A. 1.6 s B. 0.4 s
C. -0.8 s D. 0.8 s

P08. An object travels 6 m forward and then 8 m farther forward along the same line. Find the total distance travelled.

A. 14 m B. 28 m
C. 7 m D. -14 m

P09. In a motion diagram, one gap is 4 cm and a later gap is 10 cm. Calculate how much the gap increased.

A. 12 cm B. 6 cm
C. 3 cm D. -6 cm

P10. Repeated readings are 22, 24 and 26 cm. Calculate the range of the readings.

A. 8 cm B. 2 cm
C. 4 cm D. -4 cm

P11. What is a motion diagram?

P12. Why is a reference object needed when describing motion?

P13. When is a particle model appropriate?

P14. How does a motion diagram show an object at rest?

P15. Why must the timing of images be known?

P16. A motion diagram shows positions 0, 6, 12 and 18 m at equal times. What is the spacing between successive positions?

P17. A motion diagram shows positions 0, 8, 16 and 24 m at equal times. What is the spacing between successive positions?

P18. A camera records 6 successive positions separated by 0.2 s. Calculate the time from the first image to the last.

P19. A camera records 7 successive positions separated by 0.2 s. Calculate the time from the first image to the last.

P20. An object travels 9 m forward and then 12 m farther forward along the same line. Find the total distance travelled.

P21. An object travels 12 m forward and then 16 m farther forward along the same line. Find the total distance travelled.

P22. In a motion diagram, one gap is 6 cm and a later gap is 15 cm. Calculate how much the gap increased.

P23. In a motion diagram, one gap is 8 cm and a later gap is 20 cm. Calculate how much the gap increased.

P24. Repeated readings are 23, 25 and 27 cm. Calculate the range of the readings.

P25. Repeated readings are 24, 26 and 28 cm. Calculate the range of the readings.

P26. Sketch four successive positions for an object moving right at constant speed. State the time-interval assumption.

P27. Sketch a motion diagram for an object slowing down while moving right.

P28. A passenger says a bag on a train seat is stationary; a platform observer says it is moving. Explain.

P29. Why can a blurred photograph alone be insufficient for a quantitative motion description?

P30. Two diagrams have equal dot gaps but different time intervals between dots. Must their speeds be equal? Explain.